

Lecture 8

Announcement: HW 3 visible, due Oct 8 at 11:59 PM

Last time:

- Introduced expectation (for discrete and continuous r.v.s)
- Law of The unconscious statistician

Recap:

- Expectation of a discrete r.v. (if it exists): $E[X] = \sum_k x_k P(x_k)$
- "continuous": $E[X] = \int_{\mathbb{R}} x f(x) dx$

- LOTUS: $E[g(X)] = \sum_k g(x_k) P(x_k)$ (if discrete and expectation of $g(X)$ exists)
- $E[g(X)] = \int_{\mathbb{R}} g(x) f(x) dx$ ("continuous")

Notes on the above:

- In general, $E[g(X)] \neq g(E[X])$ e.g., $E[\frac{1}{X}] \neq \frac{1}{E[X]}$

- All of this extends to $X = (X_1, \dots, X_d)$
 $\uparrow$ random vector
 "multivariate r.v."

If $E[X_i]$ exists for all $i \in \{1, \dots, d\}$, then

$$E[X] = (E[X_1], \dots, E[X_d])$$

- Relationship between indicator functions and expectation

$$\mathbb{1}\{A\} = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{if } A^c \text{ occurs} \end{cases}$$

← Indicator variable

$$f(x) = cx \mathbb{1}\{0 < x < 1\}$$

$$E[\mathbb{1}\{A\}] = 1 \times P(A) + 0 \times P(A^c) = P(A)$$

- If X and Y are independent r.v.s then for all functions g, h , we have that

$$E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

e.g., Suppose A and B are independent events

$$\text{Then } E[\mathbb{1}\{A \cap B\}] = P(A)P(B)$$

How to show this?

$$\textcircled{1} \text{ use the previous fact } E[\mathbb{1}\{A \cap B\}] = P(A \cap B) = P(A)P(B)$$

$$\begin{aligned} \textcircled{2} E[\mathbb{1}\{A \cap B\}] &= E[\mathbb{1}\{A\} \mathbb{1}\{B\}] \stackrel{\text{fact}}{=} E[\mathbb{1}\{A\}] E[\mathbb{1}\{B\}] \\ &= P(A)P(B) \stackrel{\text{other fact}}{=} E[\mathbb{1}\{A\} \mathbb{1}\{B\}] \\ &= \begin{cases} 1 & \text{if } A \cap B \\ 0 & \text{o/w} \end{cases} \end{aligned}$$

useful:

$$\begin{aligned} \mathbb{1}\{A \cap B\} &= \mathbb{1}\{A\} \mathbb{1}\{B\} \\ \mathbb{1}\{A \cup B\} &= \mathbb{1}\{A\} + \mathbb{1}\{B\} - \mathbb{1}\{A \cap B\} \end{aligned}$$

DEF: $E[X]$ is referred to the first moment of X

In general, $E[X^n]$ is referred to as the n^{th} -moment of X

Also, $E[(X - E[X])^n]$ is referred to as the n^{th} central moment of X

Properties of expectation

- $E[1_A] = P(A)$
- $E[h(X)g(Y)] = E[h(X)]E[g(Y)] \quad \forall g, h$ if X and Y are independent
- Linearity of expectation: if a, b are constants, then $E[aX + b] = aE[X] + b$

↳ In general, $E[\sum_{i=1}^n a_i X_i + b] = \sum_{i=1}^n a_i E[X_i] + b$ (true for any r.v.)

Proof: (simple case, discrete r.v.)

$$E[aX + b] \stackrel{\text{lots}}{=} \sum_k (ax_k + b)p(x_k) = \underbrace{a \sum_k x_k p(x_k)}_{E[X]} + \underbrace{b \sum_k p(x_k)}_1 = aE[X] + b$$

- If there exist constants such that $P(a \leq X \leq b) = 1$, then $a \leq E[X] \leq b$
reason: The average is between the min and the max

Expectations in terms of cdfs.

- If X is discrete, $E[X] = \sum_{i=1}^{\infty} P(X \geq i) = \sum_{i=1}^{\infty} (1 - P(X < i))$

for this to be true, X can only take values in $\{0, 1, 2, \dots\}$

- If X is continuous, $E[X] = \int_0^{\infty} P(X \geq t) dt = \int_0^{\infty} (1 - F(t)) dt$

for this to be true, $X \geq 0$

(There are extensions of the above for $X \in \mathbb{R}$ or $X \in \mathbb{Z}$)

Proof

① Discrete case: $E[X] = \sum_{n=0}^{\infty} n P(X=n)$

By definition

$$= \sum_{n=1}^{\infty} n P(X=n)$$

$$0 \cdot P(X=0) = 0$$

$$= \sum_{n=1}^{\infty} \sum_{i=1}^n P(X=n)$$

$$\sum_{i=1}^{\infty} P(X=i) = n P(X=i)$$

$$= \sum_{i=1}^{\infty} \sum_{n=i}^{\infty} P(X=n)$$

(interchange sums)

$$= P(X \geq i)$$

② Continuous case: $E[X] = \int_0^{\infty} x f(x) dx$

$$= \int_0^{\infty} \left(\int_0^x 1 dy \right) f(x) dx$$

$$= \int_0^{\infty} \left(\int_y^{\infty} f(x) dx \right) dy = \int_0^{\infty} \left(1 - \int_0^y f(x) dx \right) dy = \int_0^{\infty} (1 - F(y)) dy$$

$$\int_0^\infty \left(\int_0^x 1 \, dy \right) f(x) \, dx = \int_0^\infty \int_0^\infty \mathbb{1}\{y \leq x\} f(x) \, dx \, dy$$

$$= \int_0^\infty \int_y^\infty \mathbb{1}\{y \leq x\} f(x) \, dx \, dy$$

(LOTUS for multivariate case)

• If X and Y have joint pmf, $p_{X,Y}(x,y)$, then $\mathbb{E}[g(X,Y)] = \sum_y \sum_x g(x,y) p_{X,Y}(x,y)$

" " pdf $f_{X,Y}(x,y)$ " $\mathbb{E}[g(X,Y)] = \int_{\mathbb{R}} \int_{\mathbb{R}} g(x,y) f_{X,Y}(x,y) \, dx \, dy$

↳ What if $g(X,Y) = X$?

$$\mathbb{E}[g(X,Y)] = \mathbb{E}[X] = \int_{-\infty}^\infty \int_{-\infty}^\infty x f_{X,Y}(x,y) \, dy \, dx$$

$$= \int_{-\infty}^\infty x \underbrace{\int_{-\infty}^\infty f_{X,Y}(x,y) \, dy}_{f_X(x)} \, dx$$

$$= \int_{-\infty}^\infty x f_X(x) \, dx = \mathbb{E}[X] \quad \checkmark$$

Example: Suppose the location of an accident, X , is uniformly distributed on a road of length L . At the time of the accident, suppose an ambulance is at location Y , where Y is also uniformly distributed on the road (of length L). Assume that X and Y are independent. Find $\mathbb{E}[|Y-X|]$

distance between the accident + where the ambulance is

$$\mathbb{E}[|X-Y|] = \int_{\mathbb{R}} \int_{\mathbb{R}} |x-y| f_{X,Y}(x,y) \, dy \, dx$$

$$= \int_{\mathbb{R}} \int_{\mathbb{R}} |x-y| f_X(x) f_Y(y) \, dy \, dx$$

$$= \int_{\mathbb{R}} \int_{\mathbb{R}} |x-y| \cdot \frac{1}{L} \mathbb{1}\{0 < x < L\} \cdot \frac{1}{L} \mathbb{1}\{0 < y < L\} \, dy \, dx$$

$$= \frac{1}{L^2} \int_0^L \int_0^L |x-y| \, dy \, dx = \frac{1}{L^2} \int_0^L \left(\frac{L^2}{2} + x^2 - xL \right) \, dx = \dots = \frac{L}{3}$$

$$\underbrace{\int_0^L |x-y| \, dy}_{=} = \int_0^x (x-y) \, dy + \int_x^L (y-x) \, dy = \frac{L^2}{2} + x^2 - xL$$

$$= \int_0^L \mathbb{1}\{x \geq y\} (x-y) + \mathbb{1}\{x < y\} (y-x) \, dy$$

$$= \int_0^L \mathbb{1}\{y \leq x\} (x-y) \, dy + \int_0^L \mathbb{1}\{x < y\} (y-x) \, dy$$

$$= \int_0^x (x-y) \, dy + \int_x^L (y-x) \, dy$$

• Fact: previously, learned about binomial r.v.s

If $X \sim \text{Binomial}(n, p)$

is distributed as $\text{Binomial}(n, p)$

$X \sim \text{Unif}(0, L)$
 $\frac{1}{L} \mathbb{1}_{\{0 \leq x \leq L\}}$

$X \sim \text{Binomial}(n, p)$ means that $P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$

$Y \sim \text{Binomial}(m, q)$ $P(Y=y) = \binom{m}{y} q^y (1-q)^{m-y}$

Fact: If $X \sim \text{Binomial}(n, p)$ Then X has the same distribution as $X_1 + \dots + X_n$ where the X_i 's are independent Bernoulli(p) r.v.s

$X_i \sim \text{Bernoulli}(p)$ means that $P(X=1) = p$ $E[X_i] = p \cdot 1 + (1-p) \cdot 0$
 $P(X=0) = 1-p$

Takeaway: $X \stackrel{d}{=} \sum_{i=1}^n X_i$

equal in distribution / has the same distribution

$$\Rightarrow E[X] = E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n p = np = \sum_{k=0}^n k \binom{n}{k} p^k (1-p)^{n-k}$$

See announcement on courseworks!

(*) example: A group of N people throw their hats into the center of the room and then each person chooses a hat at random. What is the expected number of people who select their own hat?

X = number of people who select their own hat

$$= \sum_{i=1}^N X_i \quad X_i \sim \text{Bernoulli}\left(\frac{1}{N}\right)$$

whether the i^{th} person got back their own hat

$$E[X] = \sum_{i=1}^N E[X_i] = N \cdot \frac{1}{N} = 1$$

Variance

Before: expected value is "center of mass" of a distribution

Now: variance is the "spread of a distribution" more precisely "fluctuation of a r.v. around its mean"

e.g., Consider two r.v.s

$$X = 0 \text{ wp } 1$$

$$Y = \begin{cases} -2 & \text{wp } 1/3 \\ -1 & \text{wp } 1/6 \\ 1 & \text{wp } 1/6 \\ 2 & \text{wp } 1/3 \end{cases}$$

Exercise

1) Show that $E[X] = E[Y]$

2) Compute and compare $\text{Var}(X)$ and $\text{Var}(Y)$

DEF: The variance of X , denoted by $\text{Var}(X)$, is defined as follows:

$$\text{Var}(X) = E[(X - E[X])^2]$$

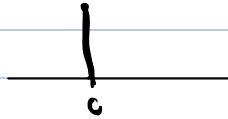


Facts about variance

- $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

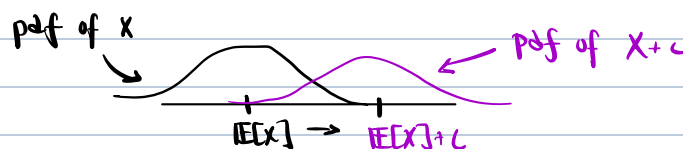
proof: exercise idea: do it for discrete case

- If there is a constant such that $\mathbb{P}(X=c) = 1$, Then $\text{var}(X) = 0$



$$\begin{aligned}\text{Var}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\ &= c^2 - (c)^2 \\ &= 0\end{aligned}$$

- If c is a constant, then $\text{var}(X+c) = \text{var}(X)$



Putting these together
 $\text{var}(aX+b)$ $a, b \in \mathbb{R}$

$$= a^2 \text{var}(X)$$

- If c is a constant, then $\text{var}(cX) = c^2 \text{var}(X)$
- If $X_1, \dots, X_n$ are mutually independent r.v. $\text{var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{var}(X_i)$

Not true in general

Important: define the sample mean by $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

Assume that $X_1, \dots, X_n$ are iid (iid means independent and identically distributed or $X_1, \dots, X_n$ are mutually independent and $X_1 \stackrel{\text{identically distributed}}{=} X_2 \stackrel{\text{identically distributed}}{=} \dots \stackrel{\text{identically distributed}}{=} X_n$)

- $\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] \stackrel{\text{identically distributed}}{=} \frac{1}{n} \cdot n \mathbb{E}[X_1] = \mathbb{E}[X_1]$

- $\text{var}(\bar{X}) = \text{var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{var}\left(\sum_{i=1}^n X_i\right) \stackrel{\text{independence}}{=} \frac{1}{n^2} \sum_{i=1}^n \text{var}(X_i) \stackrel{\text{identically distributed}}{=} \frac{1}{n^2} \cdot n \text{var}(X_1) = \frac{\text{var}(X_1)}{n}$

- DEF: The standard deviation of a random variable, X , is $\sqrt{\text{var}(X)} = \text{SD}$
- Consider minimizing $g(\theta) = (X - \theta)^2$ Then $\min_{\theta} \mathbb{E}[g(\theta)] = \min_{\theta} \mathbb{E}[(X - \theta)^2] = \text{var}(X)$

$$\Rightarrow \arg\min_{\theta} \mathbb{E}[g(\theta)] = \mathbb{E}[X]$$

Another way to write this: $\mathbb{E}[(X - \theta)^2] \geq \mathbb{E}[(X - \mathbb{E}[X])^2] \quad \forall \theta$

- Another fact: $\mathbb{E}[|X - \theta|] \geq \mathbb{E}[|X - \text{median}(X)|] \quad \forall \theta$